TSA 2011

Problem Solving

Questions taken from the TSA Oxford 2011 Section 1 question paper

Questions	Total marks	Time limit
25	25	45 minutes

Instructions

- You have **45 minutes** to answer all **25 questions**.
- Each question is worth **one mark**. There are no penalties for incorrect answers, so attempt every question.
- Each question has five options, A to E. Choose the one option you consider correct.
- Calculators are **not** permitted. You may use this paper for rough working.
- Questions are numbered 1 to 25 for this set. The number in brackets shows the question's number in the original TSA 2011 paper, e.g. **1 (Q3)**.

How the time limit was set

The time limit follows the TARA Problem Solving module (22 questions in 40 minutes), scaled to the 25 questions in this set: 25 × 40 ÷ 22 = 45.45 minutes, rounded to the nearest minute (45 minutes).

Score: _____ / 25

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1 (Q2) To play a football pools game, participants must select four matches from the fifty on the coupon. Points are scored depending on the result of each match as follows:

If the home team wins, the participant scores 1.0 point

If the away team wins, the participant scores 1.5 points

If the match is a draw, the participant scores 3.0 points

Which of the following total scores for the four matches is **not** possible?

A 8.5 points
B 9.0 points
C 9.5 points
D 10.0 points
E 10.5 points

2 (Q6) To encourage themselves to lose weight, Alma, Brigit, Cara, Danni and Evelyn each put £10 into a pot. They weighed themselves when they began and again at the end of each week. The one who had lost most weight after 4 weeks won the £50. The following chart shows their progress:

	Alma	*Brigit*	*Cara*	*Danni*	*Evelyn*
Starting weight	71kg	67kg	63½kg	66½kg	69kg
Weight after week 1	67kg	68kg	62kg	67kg	66kg
Weight after week 2	63½kg	66½kg	59kg	62kg	63½kg
Weight after week 3	61kg	66kg	58kg	61½kg	62kg
Weight after week 4	64kg	62kg	57½kg	60kg	61kg

Who won the £50?

A Alma
B Brigit
C Cara
D Danni
E Evelyn

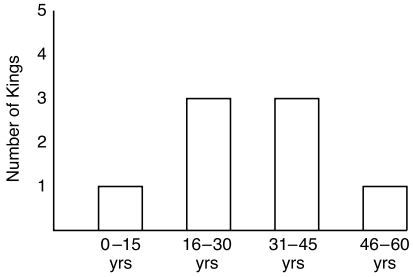
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3 (Q7) Below is a list of the Angevin and Plantagenet kings of England and the dates of their reigns.

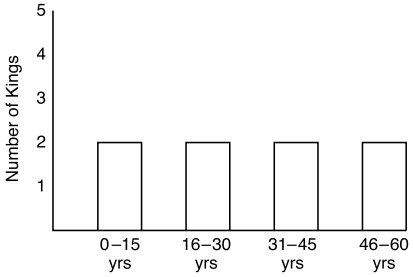
Monarch	Reign
Henry II	1154 - 1189
Richard I	1189 - 1199
John	1199 - 1216
Henry III	1216 - 1272
Edward I	1272 - 1307
Edward II	1307 - 1327
Edward III	1327 - 1377
Richard II	1377 - 1399

Which of the following bar charts represents a distribution of the lengths of the reigns of these kings?

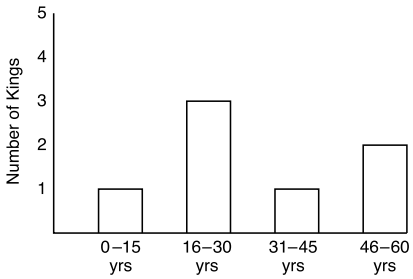
A



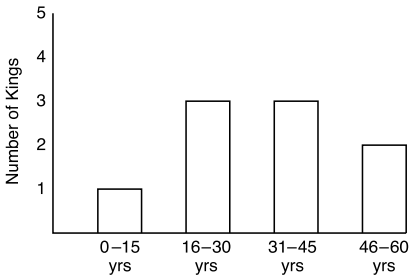
B



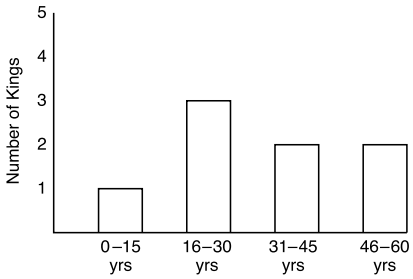
C



D



E

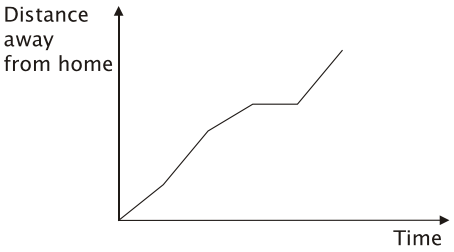


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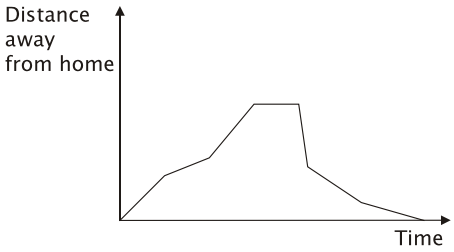
4 (Q8) A man drives off from his house and travels through the village at a constant speed. As he leaves the village behind, his speed increases to a new constant value which he maintains until he reaches a steep hill which slows him down. At the top of the hill the driver stops for a few minutes to admire the view. He realises that he has left his picnic basket at home and so drives rapidly back along his route to collect it.

Which one of the following graphs best represents the driver's distance from home throughout his journey?

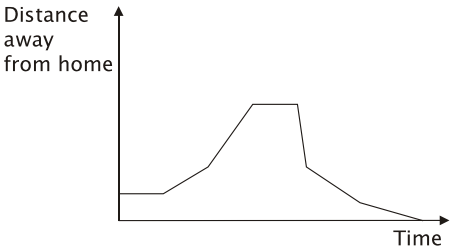
A



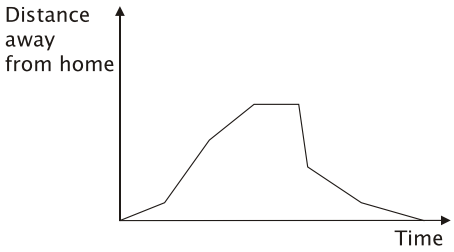
B



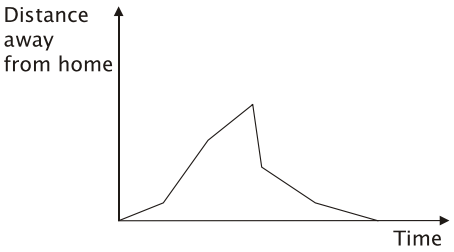
C



D



E



5 (Q12) Jars of Bestcaff coffee weighing 200g are normally £2.50 at Kostless and Savemore, but both supermarkets have a special offer this week:

KOSTLESS BESTCAFF 80p OFF	SAVEMORE BESTCAFF 25% EXTRA FREE

By how much per kilogram is Kostless's offer better value for money than Savemore's?

A £1.50 per kg
B £4.00 per kg
C £7.00 per kg
D £8.50 per kg
E £16.00 per kg

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6 (Q13) The table shows the areas of the countries of the European Union.

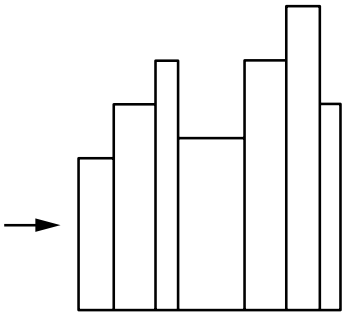
Country	*area (sq km)*	*% of EU area*	*population density (people/sq km)*
Austria	83858	1.90	99
Belgium	30510	0.70	344
Bulgaria	110912	2.50	70
Cyprus	9250	0.20	84
Czech Republic	78866	1.80	131
Denmark	43094	1.00	126
Estonia	45226	1.00	29
Finland	337030	7.60	16
France	643548	14.60	99
Germany	357021	8.10	231
Greece	131940	3.00	84
Hungary	93030	2.10	108
Ireland	70280	1.60	60
Italy	301320	6.80	196
Latvia	64589	1.50	35
Lithuania	65200	1.50	52
Luxembourg	2586	0.10	181
Malta	316	0.00	1261
Netherlands	41526	0.90	394
Poland	312685	7.10	122
Portugal	92931	2.10	114
Romania	238391	5.40	91
Spain	504782	11.40	87
Slovakia	48845	1.10	111
Slovenia	20253	0.50	99
Sweden	449964	10.20	20
United Kingdom	244820	5.50	246
European Union	**4422773**	**100.00**	

What is the minimum number of countries needed to make up at least 50% of the total land area?

A 5
B 6
C 7
D 11
E 22

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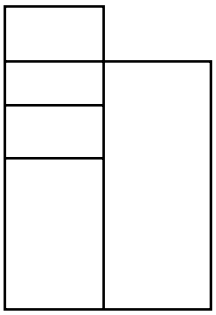
7 (Q14) There are seven books on my shelf, which are of various sizes. When I look at them straight on I see the following picture:



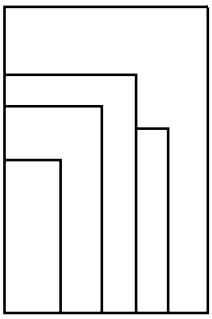
I now look at them from the left-hand end (in the direction of the arrow).

Which one of the following diagrams is **not** a possible view when seen from the left-hand end?

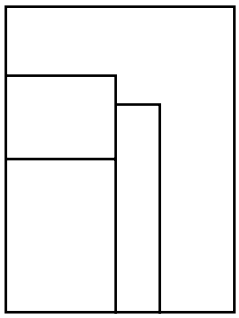
A



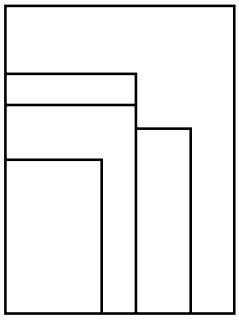
B



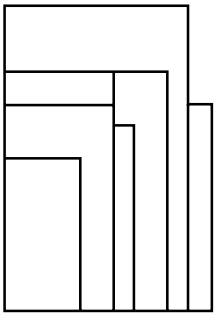
C



D



E



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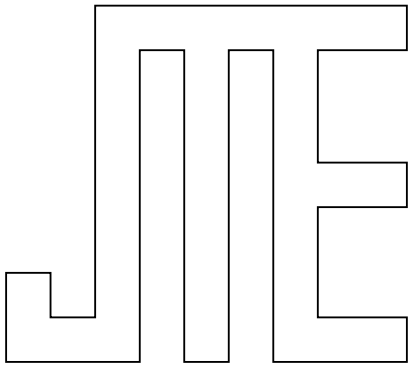
8 (Q18) Dates may be written in an eight digit form. For instance, 19 January 2005 may be written 19-01-2005.

In what year will the next date occur for which all eight digits are different?

A 2013
B 2134
C 2145
D 2345
E 2456

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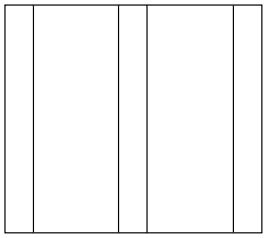
9 (Q19) This is the view from above of the large three-dimensional logo situated in the forecourt of the head office of Justin Thyme Enterprises:



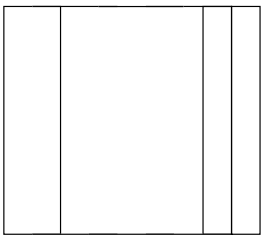
It was sculpted from a single cube of marble, and has a uniform cross-section from top to bottom.

Which of the following is **not** a possible side view of the logo?

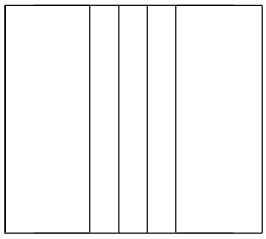
A



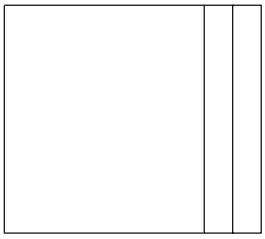
B



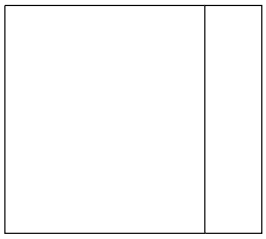
C



D



E



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10 (Q20) The table below, from a newspaper, shows how much money ($) you will need at various points in the future to buy items costing $1000 today, depending on the annual rate of inflation.

	Average yearly rate of inflation			
Years	*4 %*	*6 %*	*8 %*	*10 %*
5	1217	1338	1469	1611
6	1265	1419	1587	1772
7	1316	1504	1714	1949
8	1369	1594	1851	2144
9	1423	1690	1999	2358
10	1480	1791	2159	2594
20	2191	3207	4661	6728
30	3423	5743	10063	17449
60	10520	32988	101257	304482

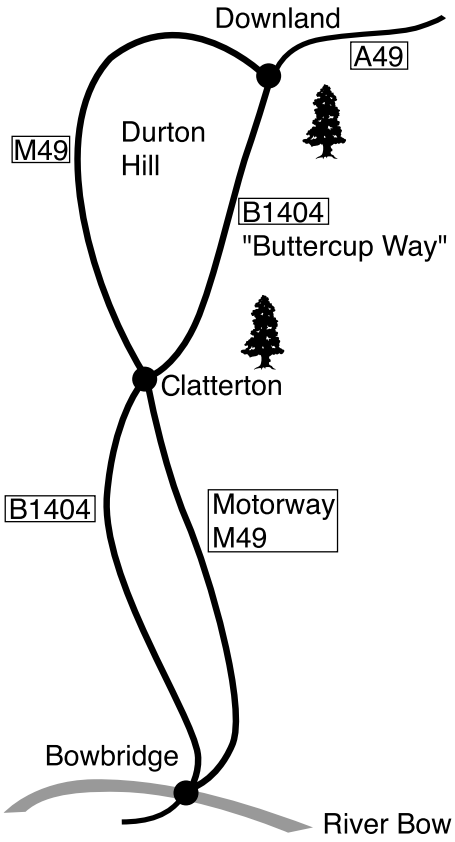
Raoul has just bought a new car for $10 000. He expects this car to last him 8 years, when he will buy another equivalent model. He puts $10 000 into a bank account ready to pay for the next car. The price of cars rises with inflation at 8% and his savings only earn interest of 6%.

How much will he need to add to his savings when he needs to buy his new car?

A $ 1612
B $ 2570
C $ 5338
D $ 5940
E $ 8510

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11 (Q24) As part of a journey, I must travel from Bowbridge to Downland. I can choose either the motorway (known locally as 'the road to hell') or the old road known as Buttercup Way. On the motorway I average 60 mph, on the old road I average 40 mph. I can change from one to the other at Clatterton where they cross.



The distances by the two roads are as follows:

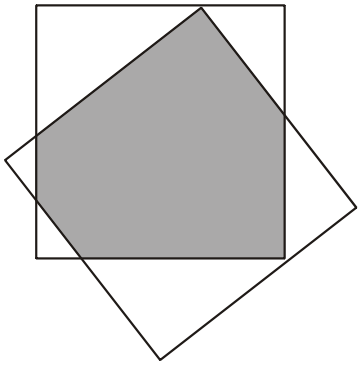
	Motorway	*Buttercup Way*
Bowbridge / Clatterton	35 miles	30 miles
Clatterton / Downland	35 miles	20 miles

If I take the fastest route, what will be the time saving over the slowest route?

A 5 min
B 10 min
C 15 min
D 20 min
E 30 min

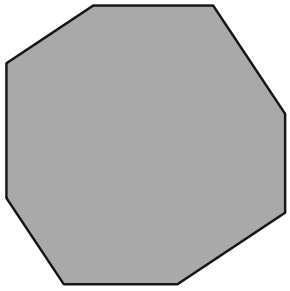
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12 (Q25) Various shapes may be made by overlapping two transparent squares, for example the dark shaded hexagon shown below:

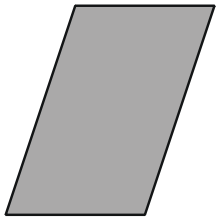


Which one of the following shapes cannot be made as the overlap of two transparent squares?

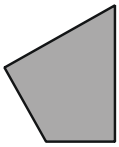
A



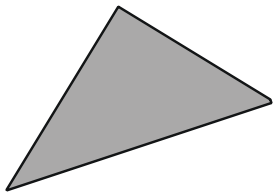
B



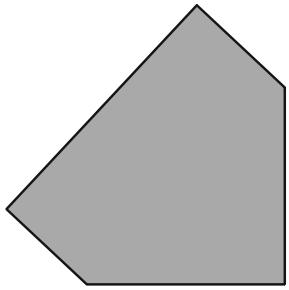
C



D



E



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13 (Q26) This table shows how household sizes in Great Britain changed between 1961 and 1991.

Percentages				
	1961	*1971*	*1981*	*1991*
Household size				
1 person	14	18	22	27
2 people	30	32	32	34
3 people	23	19	17	16
4 people	18	17	18	16
5 people	9	8	7	5
6 or more people	7	6	4	2
Total number of households (millions)	16.2	18.2	19.5	21.9
Average household size (number of people)	3.1	2.9	2.7	2.5

What was the change in the percentage of households with fewer than 3 people between 1971 and 1991?

A 7
B 8
C 10
D 11
E 17

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14 (Q30) All the teams in a local football league have now played each other once this season. Below is the current league table:

	Played	*Won*	*Drawn*	*Lost*	*Goals For*	*Goals Against*
Rovers	4	4	0	0	7	2
United	4	2	1	1	2	1
County	4	1	2	1	4	4
Wanderers	4	1	0	3	3	5
City	4	0	1	3	1	5

What was the result of the match between United and County?

A United 0 County 0

B United 1 County 0

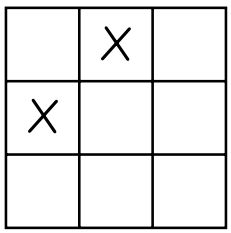
C United 0 County 1

D United 1 County 1

E United 2 County 1

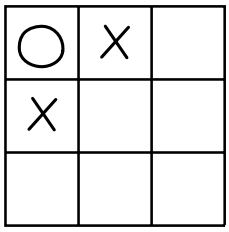
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15 (Q31) In a modified form of a popular game, each of two players at any turn may put either 'O' or 'X' as she chooses in any blank square of a 3 by 3 grid. She wins if she thereby completes a line of three identical symbols. A player is faced with this situation:

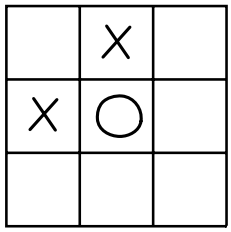


Which one of the following moves ensures that she can win at her next move?

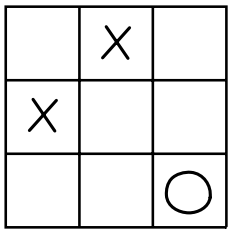
A



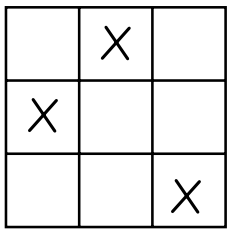
B



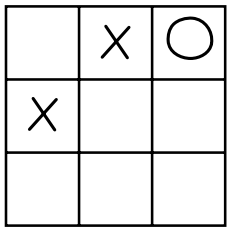
C



D



E



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16 (Q32) The following table shows the results of a test on smokers trying to quit. Of 240 people, half were given nicotine tablets and half were given a placebo (a substance containing no medication).

	Failed to quit	*Succeeded in quitting*	*Total*
Nicotine	64	56	**120**
Placebo	96	24	**120**
Total	**160**	**80**	**240**

The researchers assumed that: of the smokers given nicotine tablets, the same proportion would have quit without using them as would have quit using the placebo.

How many of the smokers that were given the nicotine tablets can the researchers conclude had quit due to taking the tablets?

A 16
B 32
C 39
D 40
E 45

17 (Q36) When my father ran a pizza shop he sold pizzas for a fixed amount plus an amount proportional to the ingredients used. A regular pizza uses twice the ingredients of a mini one and a large pizza twice those of a regular one.

Which one of the following price scales could he have used for mini, regular and large pizzas?

A £0.49, £1.00, £1.96

B £1.00, £2.00, £3.00

C £1.10, £2.20, £3.30

D £1.70, £2.00, £2.40

E £2.00, £3.00, £5.00

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18 (Q37) I often order items from a mail order catalogue. The charges for postage and packing (to be added to the order value) are as follows:

Order value:	less than £25.00	-	£4.50
	£25.00 to £49.99	-	£8.50
	£50.00 to £74.99	-	£12.00
	£75.00 to £99.99	-	£15.00
	£100.00 and over	-	£17.50

Just before sending off my latest order, I noticed that the current catalogue offers half price postage and packing for orders of 8 or more items. My order was for 7 items, with an order value of £74.60. I could not find anything else I wanted, so I just added the cheapest item in the catalogue (priced at £1.25) to my order.

By how much did adding the extra item to my order reduce the total cost?

A £3.25
B £4.50
C £4.75
D £5.75
E £6.25

19 (Q38) John wishes to have 500 copies of a brochure printed and is offered a choice of two price structures:

1. An equal cost for each brochure printed, regardless of number.
2. An initial fixed cost followed by a lower cost than structure 1 for each brochure.

Which one of the following sets of information would be sufficient for him to be able to decide which of the two price structures would be least expensive?

A The number of brochures for which the total bill would be the same by both structure 1 and structure 2.

B An example showing the cost of 500 brochures for structure 1 and 200 brochures using structure 2.

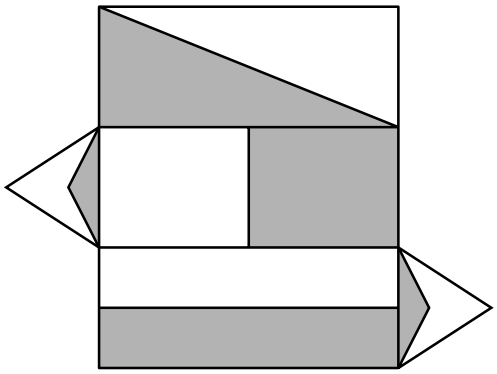
C The cost per brochure using structure 1 and the initial cost using structure 2.

D An example showing the cost of 200 brochures using structure 1 and 200 brochures using structure 2.

E The cost of 1 brochure using structure 1 and 2 brochures using structure 2.

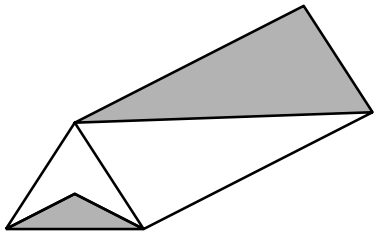
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20 (Q42) The piece of card shown below is folded to make a box with a triangular cross-section.

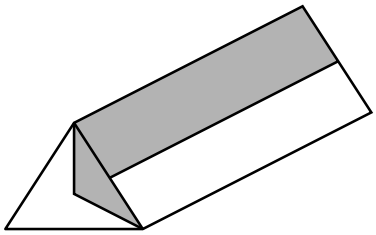


Which one of the views below can **not** be of this box?

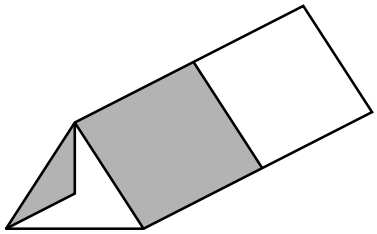
A



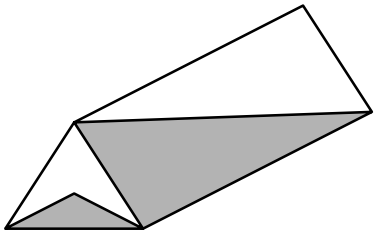
B



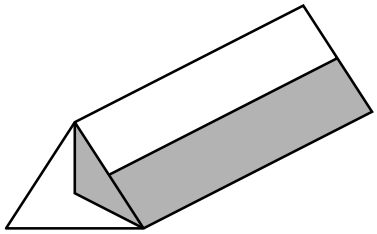
C



D



E



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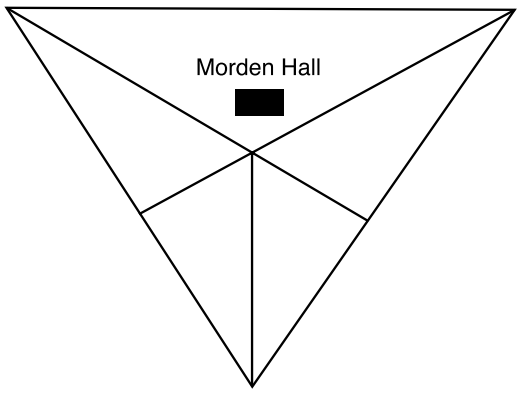
21 (Q43) At Marland College, all 120 pupils in year 11 take two languages at GCSE. The table below shows the numbers of pupils taking each language combination.

	French	*German*	*Italian*	*Spanish*
Mandarin	9	3	2	4
Spanish	14	9	13	
Italian	13	19		
German	24			

What percentage of those taking Spanish are also taking French? (Give your answer to the nearest 1% if necessary).

A 12%
B 23%
C 35%
D 39%
E 54%

22 (Q44) The diagram below shows Morden Hall and the network of paths within the grounds.



When the owner takes his dog for a walk he always starts and finishes at the intersection outside the front door and (keeping to the paths) walks one complete triangular route.

How many different routes does he have to choose from? (Clockwise and anticlockwise around the same triangle counts as one route only.)

A 5
B 7
C 10
D 11
E 12

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23 (Q48) Megan pays a subscription to Probe Information Services. This is an internet service which allows her to access up-to-date information relevant to her job. The leaflet which came with her application form is reproduced below.

Probe Information Services - Serving you in the Internet Age *Minimum monthly charge $5 - this gives access to all services and a $5 credit which will be set against the actual use made*		
Information category	*Cost per enquiry*	*Average access and download time*
Financial Markets (updated every 20 minutes)	10c	30 seconds
Exchange Rates (updated hourly)	10c	30 seconds
Company News (updated twice a day)	10c	1 minute 30 seconds
Detailed Information	50c	2 minutes 30 seconds
Commodities in World Markets	10c	30 seconds
Probe's charges do not include the cost of phone calls. All calls are at local rates		

Megan's call charges are at local rates of 10c for up to 2 minutes and 5c per additional minute or part of a minute. In the first month, Megan seeks detailed information on 5 companies and uses the other services a total of 42 times.

How much will Probe bill her?

A $ 5.00
B $ 6.70
C $ 7.50
D $ 9.95
E $ 11.65

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24 (Q49) A group of Construction students must choose their specialist options from the following list:

Bricklaying, Damp-proofing, Drainage, Flooring, Joinery, Plastering, Roofing

Each student must choose three options. The available options may be combined with each other in any way, with the exception of the restrictions that Damp-proofing and Drainage together may not be combined with Bricklaying or Plastering because of timetable constraints, and that students choosing Joinery must also choose Flooring.

How many possible combinations including Drainage could the students choose from?

A 7

B 8

C 9

D 10

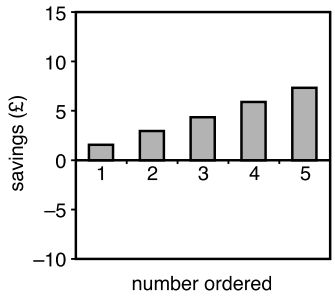
E 11

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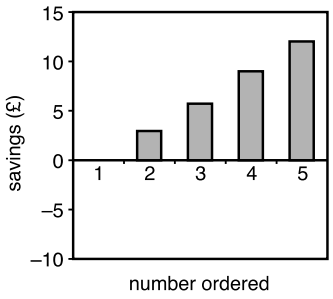
25 (Q50) I am a member of a book club. Every month I must choose at least one book. All books are sold at a discount of £2.00 on shop prices. I pay a fixed charge for postage regardless of which books I order.

Which one of the following graphs could show the relationship between the number of books ordered in a month and the total saving on shop prices (allowing for postage charge)?

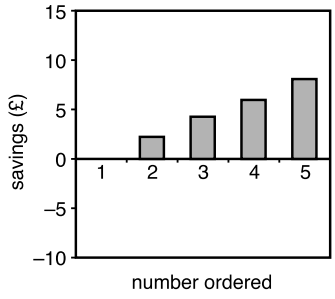
A



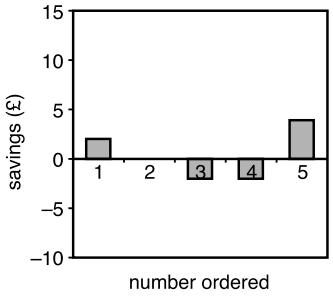
B



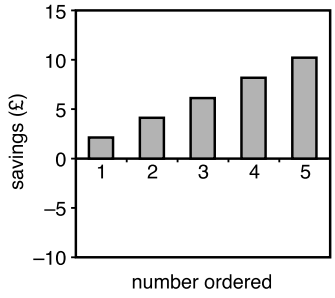
C



D



E



END OF TEST

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